

Project Title:

AWS CloudTrail Root Login Alert using CloudWatch and SNS

Project Description:

Configured a real-time AWS security monitoring system to detect **Root user login activity** using CloudTrail logs, CloudWatch metric filters, and alarms, with automated email notifications via SNS.

Architecture Overview:

Root Login
↓
CloudTrail captures event
↓
CloudWatch Logs stores logs
↓
Metric Filter detects Root login
↓
CloudWatch Alarm triggers
↓
SNS sends Email Alert

Implementation Steps:

1. Created SNS Topic and Subscription

- Created an SNS topic named root-login-alert
 - Configured email subscription
 - Verified email by confirming subscription
-

2. Configured CloudTrail

- Created a trail named security-trail
- Enabled **Management Events (Read & Write)**

- Configured S3 bucket for log storage
 - Enabled **CloudWatch Logs integration**
 - Created log group: cloudtrail-root-login-logs
 - Created IAM role: CloudTrail_CloudWatchLogs_Role
-

3. Enabled CloudWatch Logging

- Verified CloudTrail logs are delivered to CloudWatch
 - Confirmed log streams are being generated
-

4. Created Metric Filter

- Navigated to CloudWatch Log Group: cloudtrail-root-login-logs
- Created metric filter with pattern:

`{ $.userIdentity.type = "Root" && $.eventName = "ConsoleLogin" }`

- Configured:
 - Metric namespace: CloudTrailMetrics
 - Metric name: RootLoginCount
 - Metric value: 1
-

5. Created CloudWatch Alarm

- Selected metric: CloudTrailMetrics → RootLoginCount
 - Configured threshold:
 - Static threshold: ≥ 1
 - Set evaluation period: 1 datapoint
 - Attached SNS topic: root-login-alert
 - Created alarm: Root-Login-Alarm
-

6. Tested the Setup

- Logged out of AWS
 - Logged in using **Root account**
 - Verified:
 - CloudTrail captured event
 - Metric filter triggered
 - Alarm state changed to **ALARM**
 - Email notification received via SNS
-

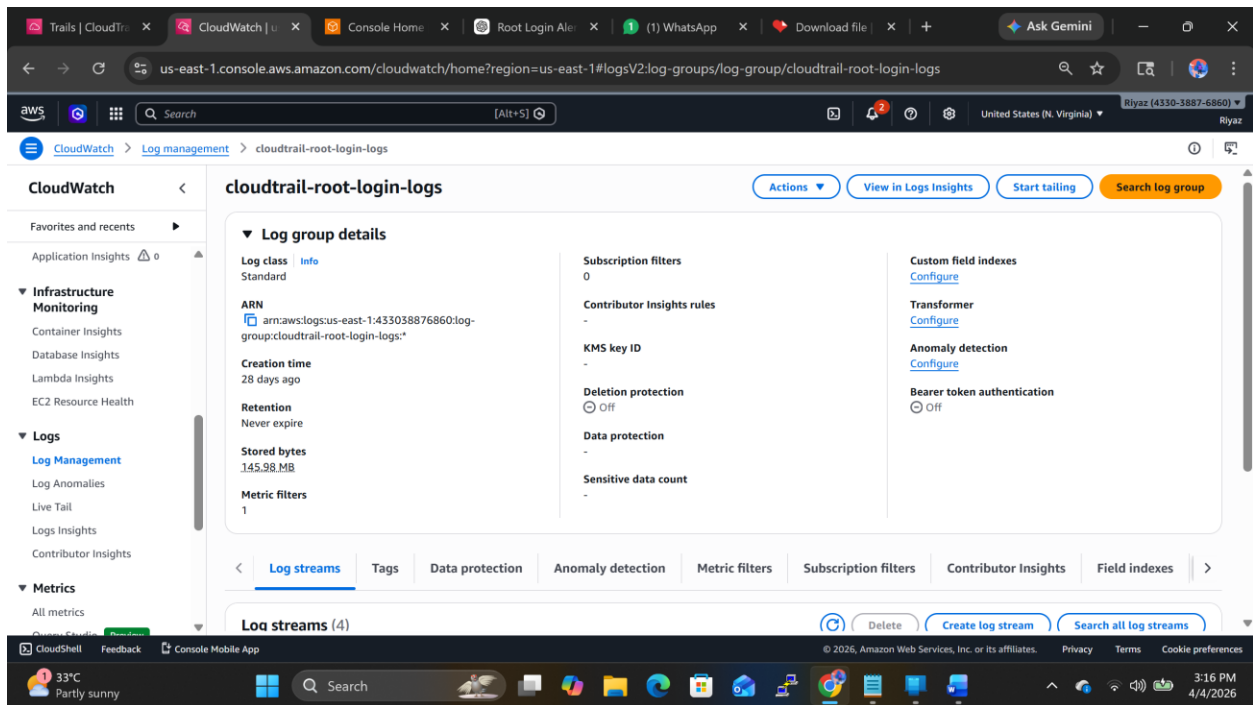
Outcome:

- Successfully implemented **real-time AWS security monitoring**
 - Enabled alerting for sensitive **Root account usage**
 - Built an automated pipeline integrating multiple AWS services
-

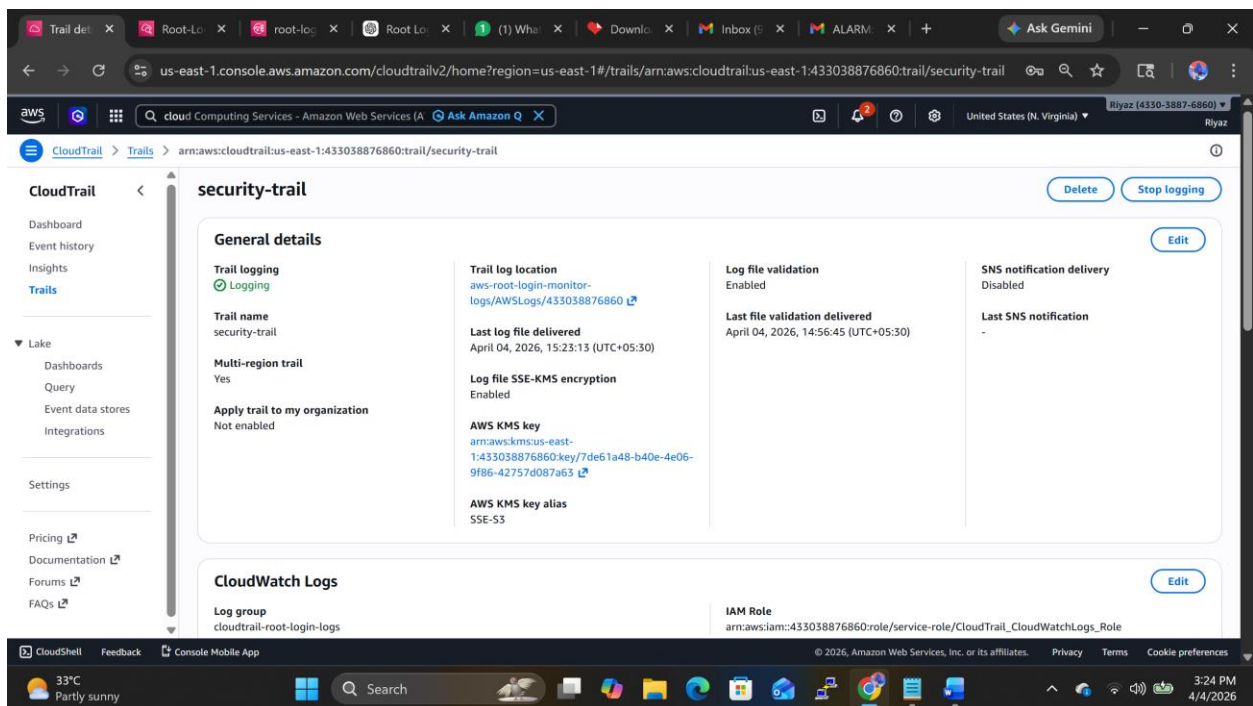
Key AWS Services Used:

- **CloudTrail** – Captures AWS account activity
- **CloudWatch Logs** – Stores and analyzes logs
- **CloudWatch Alarm** – Triggers alerts based on conditions
- **SNS (Simple Notification Service)** – Sends notifications

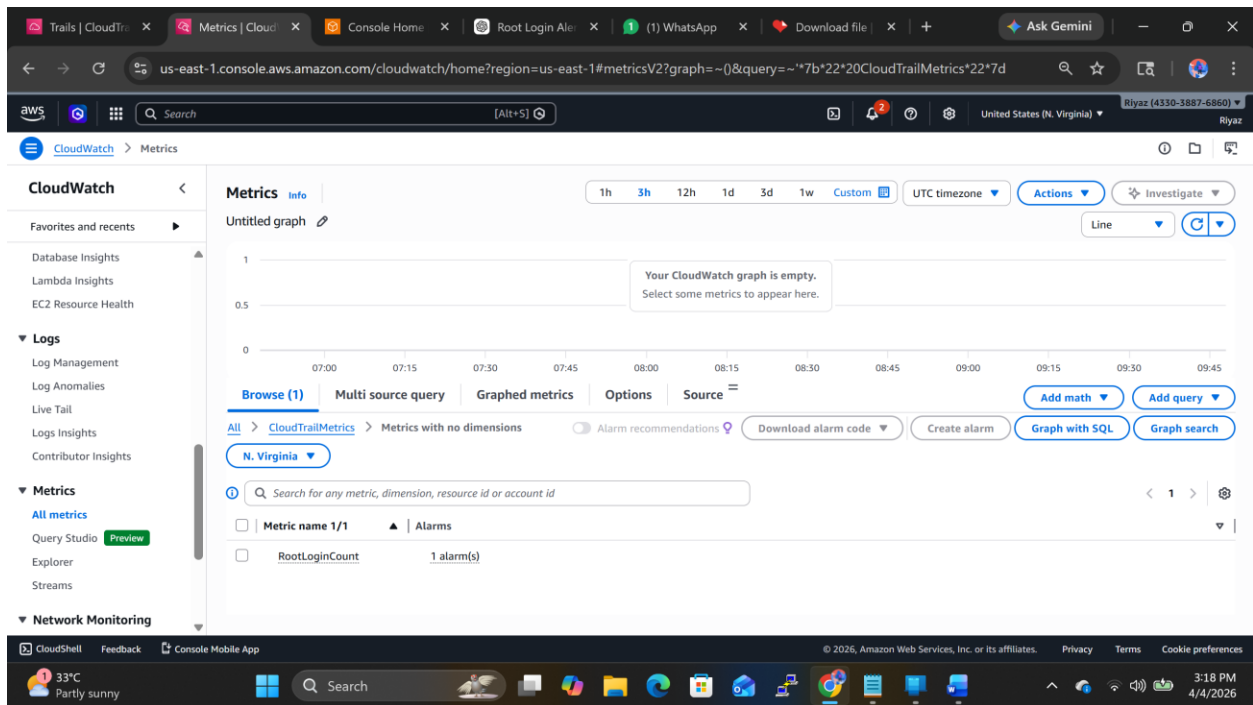
Cloudwatch log groups created:



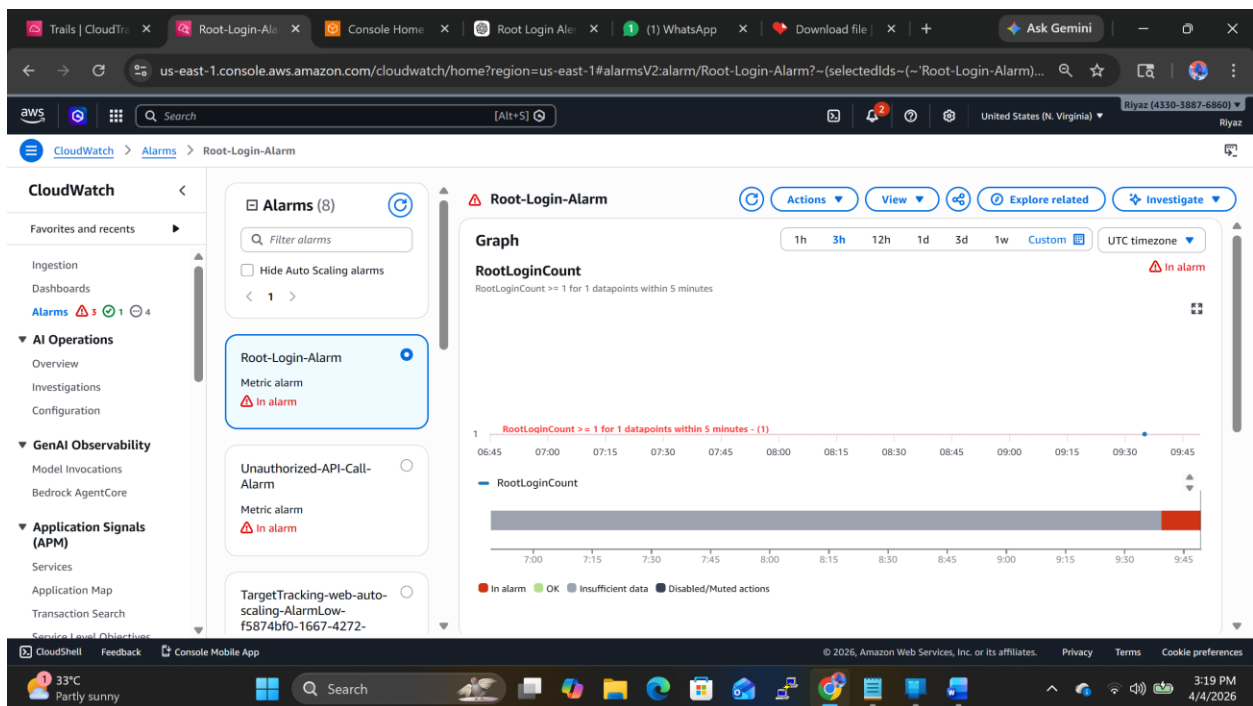
Cloudtrail created:



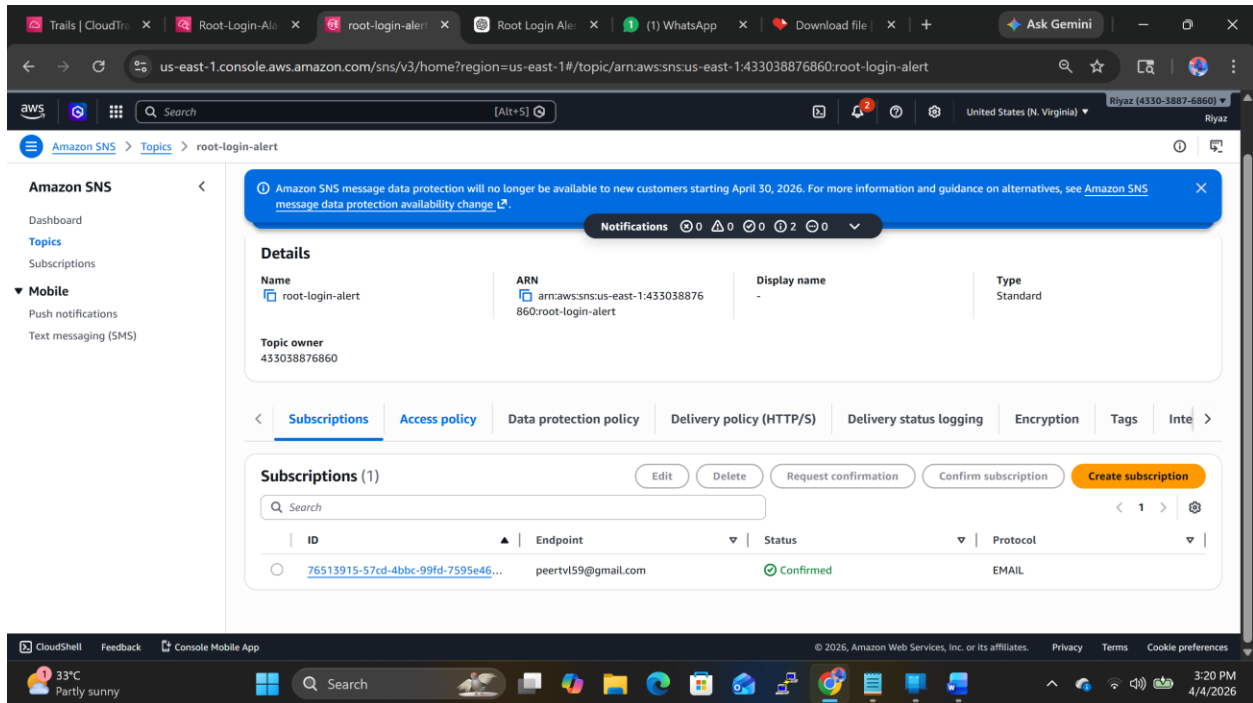
Cloudtrail metrics created:



Root login alarm created:



SNS created:



Gmail SMS while login into root account:

